



**The University of Azad Jammu and Kashmir,
Muzaffarabad**

Name: Muneeba Arshad

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Course Title: Computer Networks

Course Code: SE-2203

Lab No : 09

Submitted To: Engr. Dr. Asma Javaid

Department Of Software Engineering

✓ Switching & ARP (MAC Address)

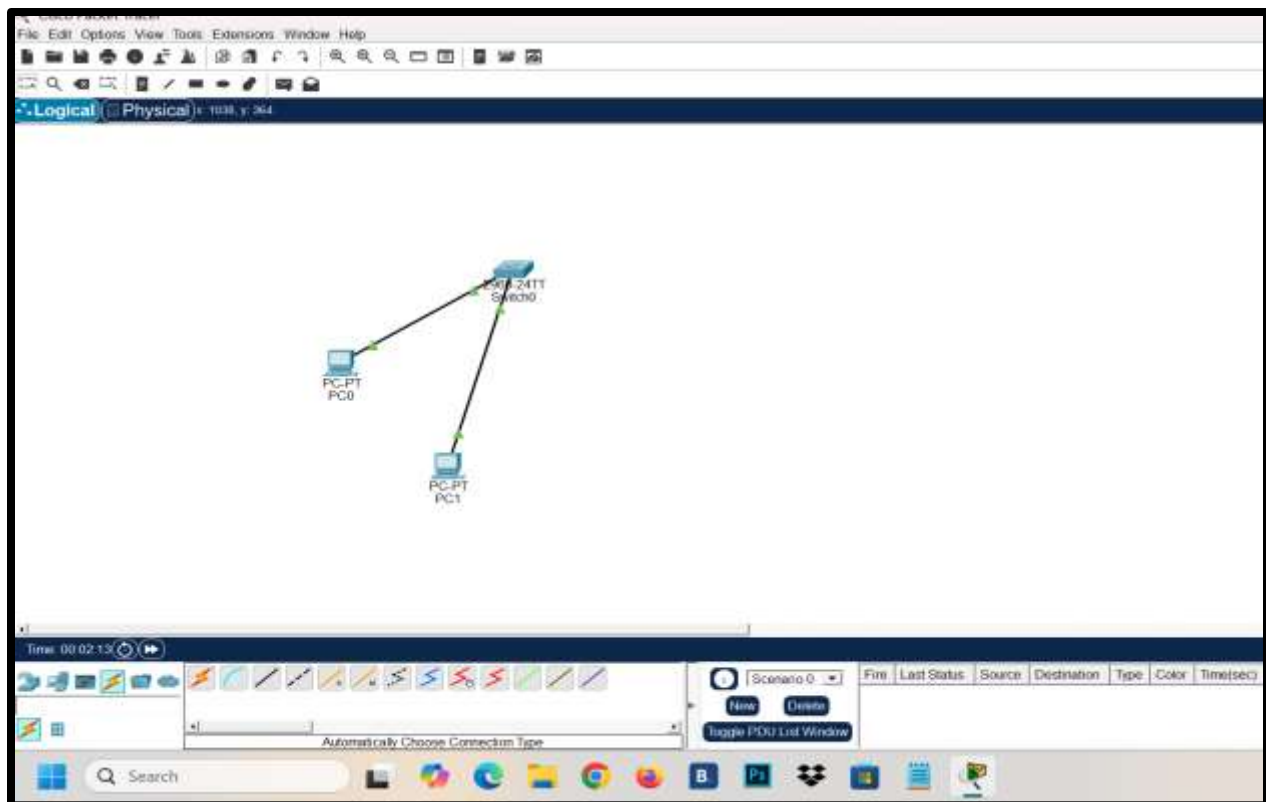
➤ Objective:

The primary objective of this lab is to understand **Layer 2 switching** and the **Address Resolution Protocol (ARP)**.

- Observe how data travels between PCs via a switch.
- Understand how a switch learns and stores MAC addresses in its **CAM table**.
- Identify the mapping between IP addresses and MAC addresses through ARP.
- Manage the switch's MAC address table using CLI commands.

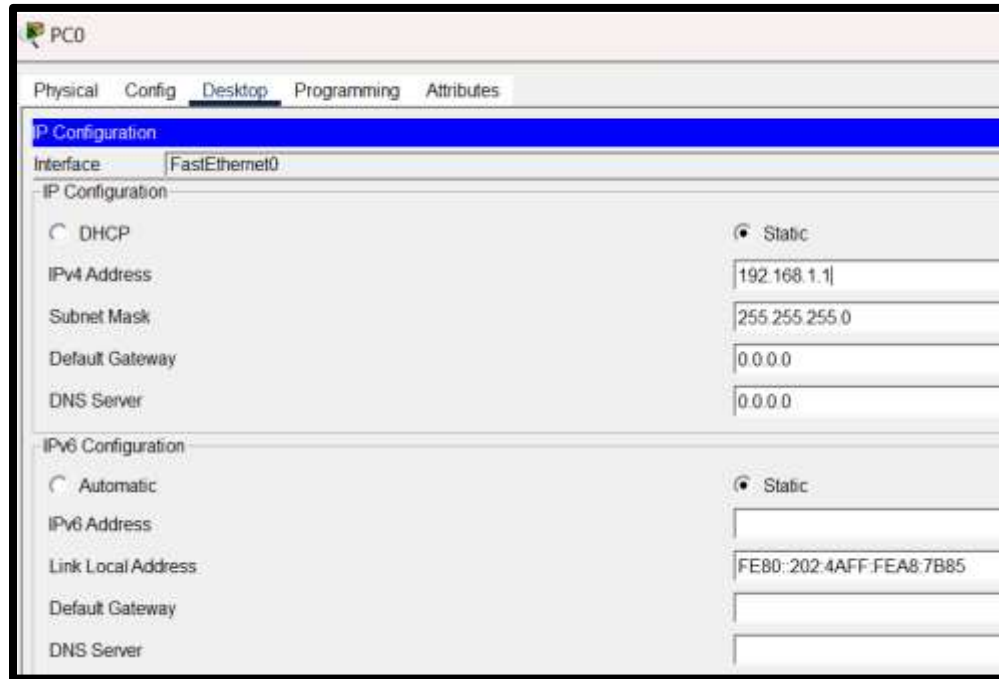
➤ Create Network:

Build a basic topology in Cisco Packet Tracer consisting of at least two PCs connected to a switch using straight-through cables.



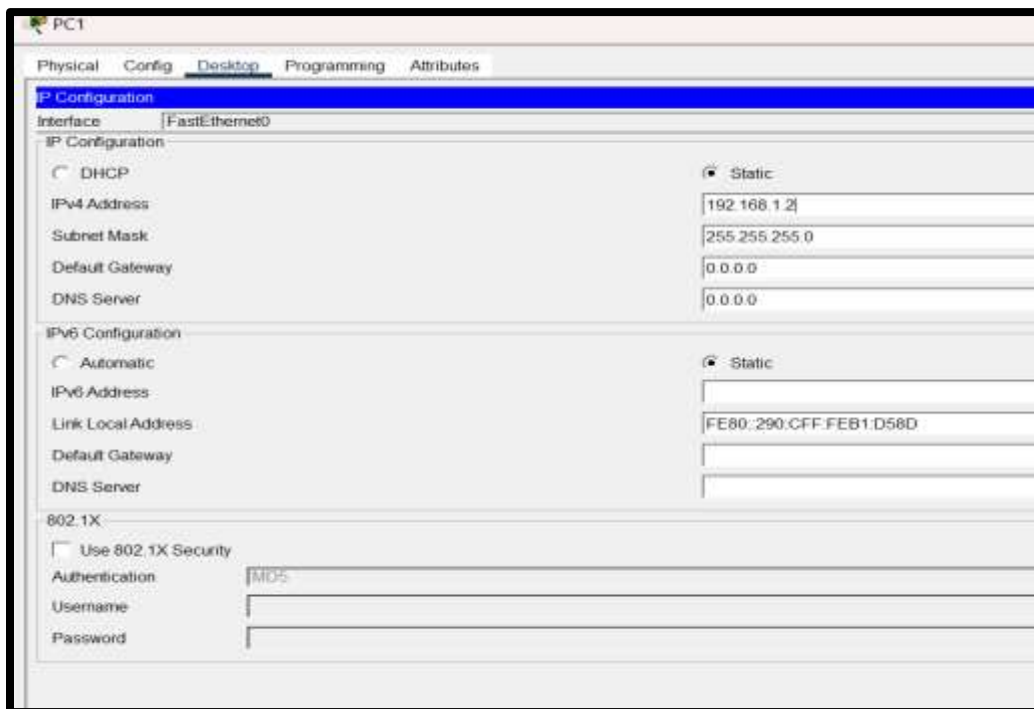
➤ Assign IP Addresses:

Each PC must be assigned a unique IP address within the same subnet to allow for communication.



The screenshot shows the configuration window for PC0. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Interface' is 'FastEthernet0'. The 'IP Configuration' section has two radio buttons: 'DHCP' (unselected) and 'Static' (selected). The 'Static' configuration fields are filled with: IPv4 Address: 192.168.1.1, Subnet Mask: 255.255.255.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. The 'IPv6 Configuration' section also has two radio buttons: 'Automatic' (unselected) and 'Static' (selected). The 'Static' configuration fields are filled with: IPv6 Address: (empty), Link Local Address: FE80::202:4AFF:FEA8:7B85, Default Gateway: (empty), and DNS Server: (empty).

IP Configuration	
Interface	FastEthernet0
☐ DHCP ☒ Static	
IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
☐ Automatic ☒ Static	
IPv6 Address	
Link Local Address	FE80::202:4AFF:FEA8:7B85
Default Gateway	
DNS Server	



The screenshot shows the configuration window for PC1. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Interface' is 'FastEthernet0'. The 'IP Configuration' section has two radio buttons: 'DHCP' (unselected) and 'Static' (selected). The 'Static' configuration fields are filled with: IPv4 Address: 192.168.1.2, Subnet Mask: 255.255.255.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. The 'IPv6 Configuration' section also has two radio buttons: 'Automatic' (unselected) and 'Static' (selected). The 'Static' configuration fields are filled with: IPv6 Address: (empty), Link Local Address: FE80::290:CFF:FEB1:D58D, Default Gateway: (empty), and DNS Server: (empty). Below the IP configuration, the '802.1X' section is visible, with a checkbox for 'Use 802.1X Security' (unchecked). The 'Authentication' section is filled with: Username: (empty), Password: (empty), and Authentication: (empty).

IP Configuration	
Interface	FastEthernet0
☐ DHCP ☒ Static	
IPv4 Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
☐ Automatic ☒ Static	
IPv6 Address	
Link Local Address	FE80::290:CFF:FEB1:D58D
Default Gateway	
DNS Server	

802.1X	
☐ Use 802.1X Security	
Authentication	
Username	
Password	

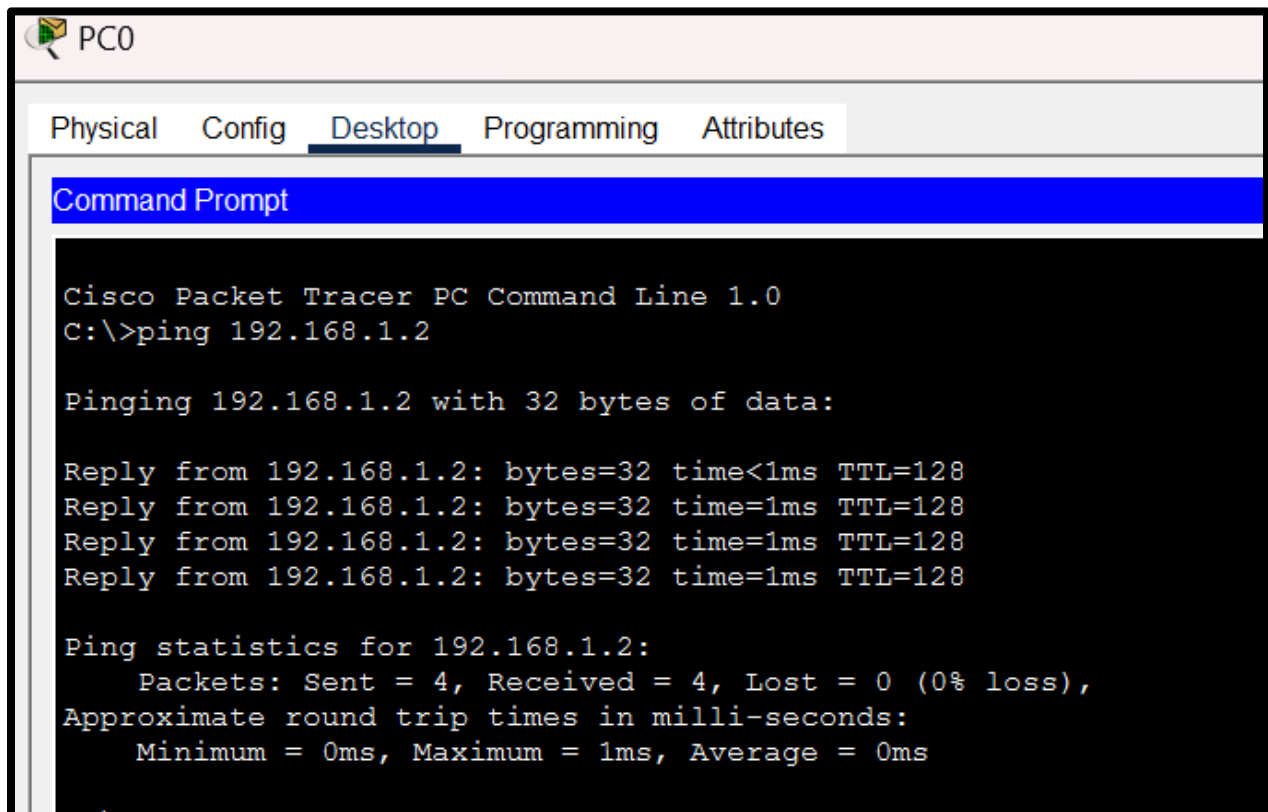
➤ Check MAC Table before Ping :

This is to verify that the table is likely empty or lacks the PC addresses before they start talking.

```
Switch>enable
Switch#
Switch# show mac address-table
          Mac Address Table
-----
Vlan      Mac Address      Type      Ports
----      -
1         0090.0cb1.d58d    DYNAMIC   Fa0/2
Switch#
```

➤ Ping between PCs:

This triggers the ARP process to resolve the destination MAC address and forces the switch to learn which MAC address is associated with which physical port.



➤ **Check MAC Table after ping :**

After the ping, you will run the show mac address-table command again. At this stage, we should see the MAC addresses of the PCs automatically populated in the switch's ta

```
Switch# show mac address-table
```

Vlan	Mac Address	Type	Ports
1	0002.4aa8.7b85	DYNAMIC	Fa0/1
1	0090.0cb1.d58d	DYNAMIC	Fa0/2

```
Switch#
```

➤ **Clear MAC Table :**

Finally, we will use the command clear mac address-table dynamic to manually wipe the switch's learned addresses. Then verify that the table has been cleared to understand how the switch handles dynamic address aging and management.

```
Switch# clear mac address-table dynamic
```

```
Switch# show mac address-table
```

Vlan	Mac Address	Type	Ports
------	-------------	------	-------

```
Switch#
```

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